LESSON 3.3

Determine Equivalent Ratios Using Tables

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a ratio table to find equivalent ratios.

WHY THIS MATTERS Brian will use the recipe shown to make clay. He needs to make multiple batches of clay. How can Brian determine how much baking soda and cornstarch he needs for multiple batches of clay to get the same consistency?

REMEMBER

Determine Equivalent Ratios Using Tables

Equivalent Ratios & Ratio Tables A ratio table lists equal ratios in rows. To make a new row, you multiply BOTH numbers by the same amount, called the scale factor.

1. Multiply or divide both terms in a ratio by the same non-zero scale factor.
2. In a ratio table, rows and columns maintain the same proportional unit rate.
3. Use additive or multiplicative patterns to find missing values in the table

Words: Ratio Tables *(Tablas de razones)* · Ratio table *(Tabla de razones)* · Equivalent ratio *(Razón equivalente)* · Scale factor *(Factor de escala)* · Pattern *(Patrón)*

1 **Circle the letter of the best answer. Show how you know.**

6 pencils cost $3.00. What is the price per pencil?

(A) $0.50 (B) $0.60

(C) $2.00 (D) $18.00

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

2 **Circle the letter of the best answer. Show how you know.**

A rate is a UNIT rate when the second quantity is what number?

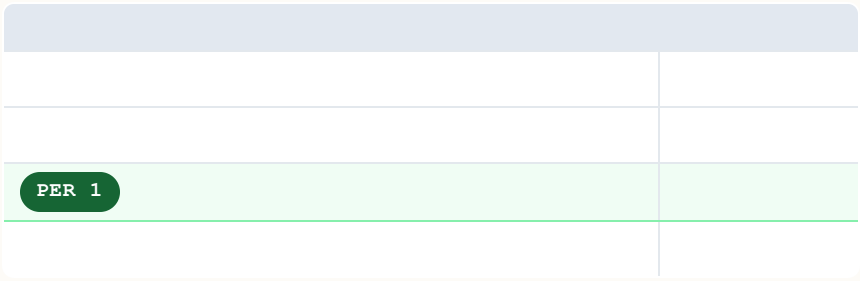
(A) 0
(B) the larger number
(C) any whole number
(D) 1

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A car travels 120 miles on 4 gallons of gas. What is the unit rate in miles per gallon?

(A) 24 miles per gallon
(B) 30 miles per gallon
(C) 116 miles per gallon
(D) 480 miles per gallon



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Which is the better buy: 6 granola bars for $3.00, or 10 granola bars for $4.50?

(A) 6 bars for $3.00
(B) They cost the same per bar
(C) 10 bars for $4.50
(D) You cannot tell

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The Chef Academy food truck makes a Berry Blast smoothie with 3 scoops of frozen berries for every 2 cups of yogurt. A customer orders enough for a party of 20 people, and each person gets one serving. The food truck needs to use a ratio table to figure out how many scoops and cups to prepare. One serving uses 3 scoops of berries and 2 cups of yogurt. How many scoops for 20 servings?

(A) 23
(B) 30
(C) 40
(D) 60

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

The Chef Academy food truck makes a Berry Blast smoothie with 3 scoops of frozen berries for every 2 cups of yogurt. A customer orders enough for a party of 20 people, and each person gets one serving. The food truck needs to use a ratio table to figure out how many scoops and cups to prepare. How many cups of yogurt are needed for 20 servings?

(A) 20
(B) 22
(C) 40
(D) 60

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

SENTENCE STARTER I found the 4-batch row by multiplying ___ and ___ by ___, which is faster than adding ___ and ___ over and over ___ times.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.3

Determine Equivalent Ratios Using Tables

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a ratio table to find equivalent ratios.

WHY THIS MATTERS Brian will use the recipe shown to make clay. He needs to make multiple batches of clay. How can Brian determine how much baking soda and cornstarch he needs for multiple batches of clay to get the same consistency?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each food truck order to its missing ratio-table value.

_____ 3 burgers cost $12, so 5 burgers cost ___

_____ Ratio table 7:2, partner of 21 is ___

_____ Ratio table 5:6, partner of 30 in the first column is ___

_____ 2 drinks cost $3, so 8 drinks cost ___

(A) $20

(B) 6

(C) 36

(D) $12

2 **Write the name of the correct group on each line.**

The food truck's lemonade ratio is 1 cup lemon juice to 5 cups water (1:5). Sort each table row by whether it keeps the ratio or breaks it.

Groups: Keeps the 1:5 Ratio | Breaks the Ratio

_____ 2:10

_____ 4:20

_____ 6:30

_____ 2:8

_____ 3:12

_____ 5:30

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

The Chef Academy food truck makes a Berry Blast smoothie with 3 scoops of frozen berries for every 2 cups of yogurt. A customer orders enough for a party of 20 people, and each person gets one serving. The food truck needs to use a ratio table to figure out how many scoops and cups to prepare. Which pair is NOT equivalent to 3:2?

(A) 5:4
(B) 6:4
(C) 9:6
(D) 30:20

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The food truck sells tacos at a steady rate: 2 tacos cost $5. At that rate, how much do 6 tacos cost?

(A) $11
(B) $12
(C) $15
(D) $30

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A smoothie recipe at the truck uses 4 cups of mango for every 3 cups of pineapple. How many cups of mango are needed for 9 cups of pineapple?

(A) 9
(B) 10
(C) 12
(D) 13

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Catch the Food Truck Error

1. **Read the order** The hot sauce mix uses a ratio of 2 spoons of pepper to 5 spoons of tomato (2:5). Find an equivalent row for a bigger batch.
2. **Pick a scale factor** To make the batch bigger, Maya picks a scale factor of 3. A scale factor should multiply BOTH parts.
3. **Scale the pepper** 2 + 1 = 3 spoons of pepper
4. **Scale the tomato** 5 + 1 = 6 spoons of tomato

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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